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Título: Projeto de Bloco: Engenharia de Softwares Escaláveis - TP3

Link da implementação:

https://github.com/guilhermecosta/INFNET/tree/master/projeto_bloco_engenharia_software_escalavel/TP3

Serviços orquestrados pelo docker compose

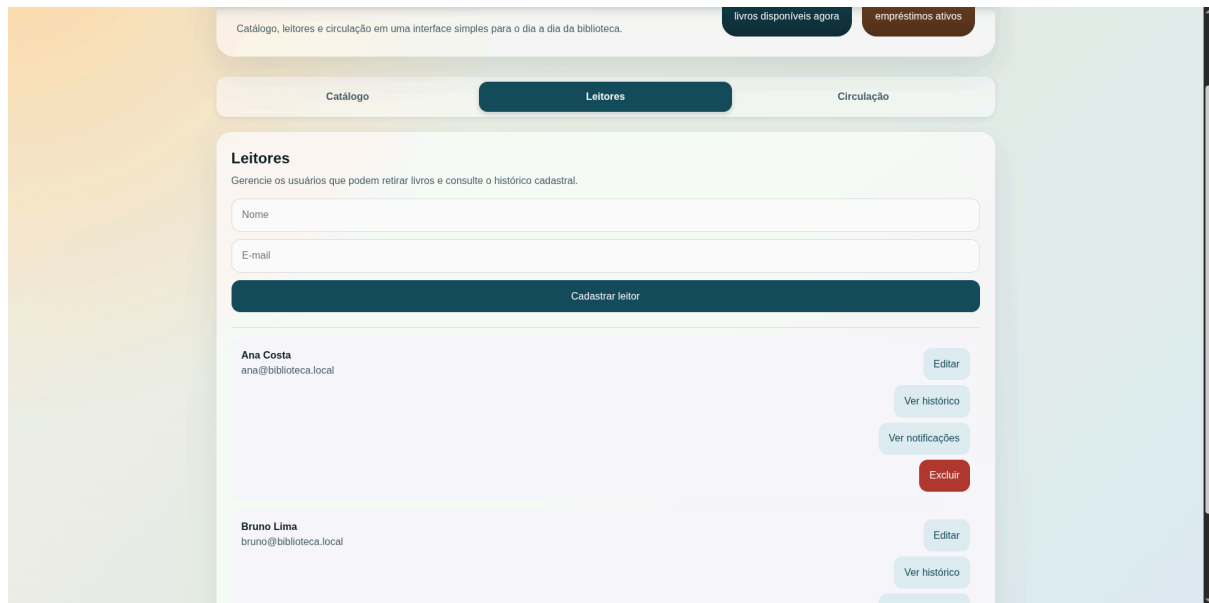
```
guichina@guichina: ~/infnet/projeto_bloco_engenharia_software_escalavel/TP3 [master [?28] [us-east-1]]
* > docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED    STATUS    PORTS
tp3-backend-1       tp3-backend         "java -jar app.jar"    backend     6 minutes ago    Up 5 minutes    0.0.0.0:8180->8080/tcp, [::]:8180->8080/tcp
tp3-frontend-1      tp3-frontend        "/docker-entrypoint..." frontend     6 minutes ago    Up 5 minutes    0.0.0.0:4173->80/tcp, [::]:4173->80/tcp
tp3-notificacoes-service-1 tp3-notificacoes-service "java -jar app.jar"    notificacoes-service 44 minutes ago    Up 5 minutes    0.0.0.0:8181->8081/tc

guichina@guichina: ~/infnet/projeto_bloco_engenharia_software_escalavel/TP3 [master [?28] [us-east-1]]
<> █
```

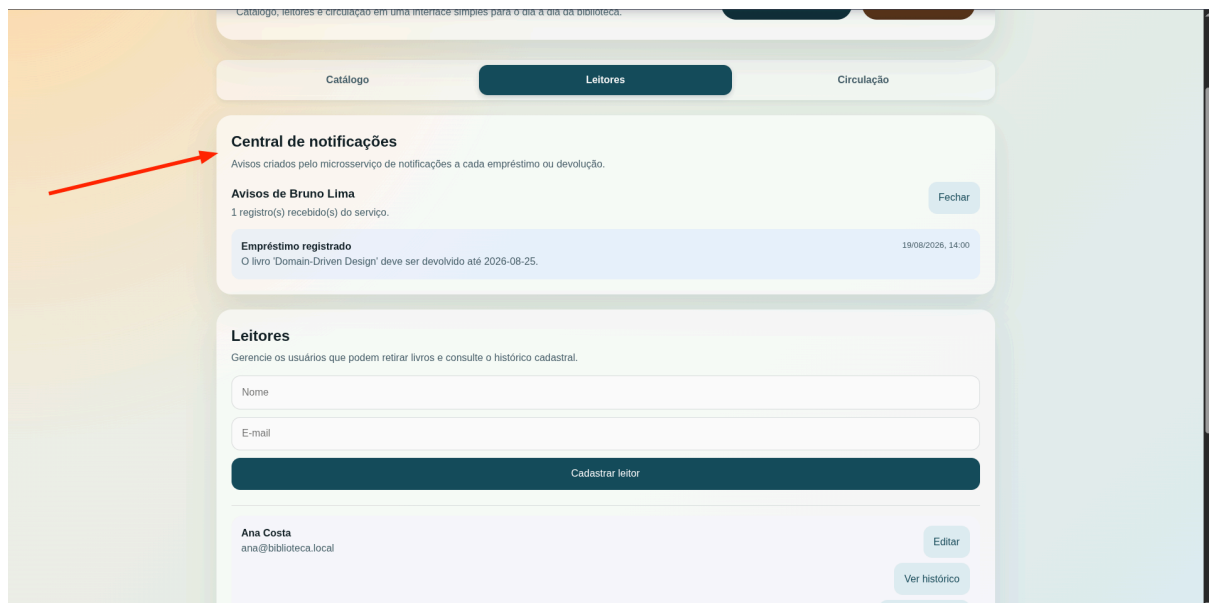
Nova interface frontend



Agora é possível “ver notificações” dos leitores. Isso está integrado ao novo microserviço de notificações



Ao registrar um empréstimo para um dado leitor, isso é mostrado nas notificações via comunicação com o novo serviço de notificações



Testes Backend

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Hibernate:
  select
    e1_0.id
  from
    emprestimos e1_0
  join
    leitores l1_0
      on l1_0.id=e1_0.leitor_id
  where
    l1_0.id=?
    and e1_0.ativo
  fetch
    first ? rows only
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.095 s -- in com.example.biblioteca.repository.RepositoryPersistenceTest
[INFO] Running com.example.biblioteca.service.EmprestimoServiceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.106 s -- in com.example.biblioteca.service.EmprestimoServiceTest
[INFO] Results:
[INFO]
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 4.040 s
[INFO] Finished at: 2026-08-19T10:39:24-03:00
[INFO]
guichina@Guichina ../infnet/projeto_bloco_engenharia_software_escalavel/TP3/backend master [787] [v0.0.1-SNAPSHOT] [v21.0.12] via [us-east-1] [4s]
```

Testes novo microserviço de notificação

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
2026-08-19T10:39:57.375-03:00 INFO 277760 --- [notificacoes-service] [main] c.e.n.r.NotificacaoRepositoryTest : Started NotificacaoRepositoryTest
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as an agent to your build as described in Mockito's documentation: https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
WARNING: A Java agent has been loaded dynamically (/home/guichina/.m2/repository/net/bytebuddy/byte-buddy-agent/1.17.5/byte-buddy-agent-1.17.5.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
Hibernate: insert into notificacoes (criada_em,leitor_id,leitor_nome,mensagem,tipos,titulo,id) values (?,?,?,?,?,?,default)
Hibernate: insert into notificacoes (criada_em,leitor_id,leitor_nome,mensagem,tipos,titulo,id) values (?,?,?,?,?,?,default)
Hibernate: insert into notificacoes (criada_em,leitor_id,leitor_nome,mensagem,tipos,titulo,id) values (?,?,?,?,?,?,default)
Hibernate: select n1_0.id,n1_0.criada_em,n1_0.leitor_id,n1_0.leitor_nome,n1_0.mensagem,n1_0.tipos,n1_0.titulo from notificacoes n1_0 where n1_0.leitor_id=? order by n1_0.criada_em desc
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.182 s -- in com.example.notificacoes.repository.NotificacaoRepositoryTest
[INFO] Running com.example.notificacoes.service.NotificacaoServiceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.198 s -- in com.example.notificacoes.service.NotificacaoServiceTest
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 3.418 s
[INFO] Finished at: 2026-08-19T10:39:58-03:00
[INFO]
guichina@Guichina ../notificacoes-service master [716] [v1.0.0] [v21.0.12] via [us-east-1] [4s]
```